

JOSHUA KIRBY

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Bilingual (English/Japanese) BSc Artificial Intelligence student on track for First Class.

Experienced in deploying production RAG architectures and full-stack AI applications.

EDUCATION

University of Birmingham

BSc Artificial Intelligence & Computer Science

Expected 2027

First Class Level (75% Average)

TECHNICAL SKILLS

- **Languages:** TypeScript, Python, SQL, JavaScript
- **Core Stack:** Next.js, React, Node.js, Tailwind CSS, Drizzle ORM
- **AI Engineering:** RAG Architectures, OpenAI API / SDKs, Vercel AI SDK, Vector DBs (Pinecone)
- **Cloud & Infra:** AWS (Lambda, DynamoDB, S3, EC2), Cloudflare Workers, SST, Serverless Architecture
- **Tools:** Git, Docker, PostHog, Vitest

EXPERIENCE

Genki Japanese and Culture School | [View Case Study](#)

Software Engineer

Fukuoka, Japan

Sep 2023 – Jun 2024

- **Sole Engineer:** Designed and deployed a production RAG chatbot for [GenkiJACS](#) and [ICNagoya](#), handling **10,000+** messages from **2,000+** users.
- **Business Impact:** Saved **~70 hours/week** by automating responses to repetitive queries, allowing staff to focus on high-value sales.
- **Pipeline Engineering:** Engineered a zero-maintenance scraping pipeline that automatically updates the Pinecone vector database when website content changes.
- **Evaluation & Optimisation:** Conducted an ablation study (700+ judgments) to validate retrieval improvements (keyword injection, query expansion), achieving a **62% baseline relevance** (vs 43% un-optimised).
- **Smart Handoff:** Engineered a handover protocol that detects critical queries and injects conversation IDs into support emails, maintaining context for staff.
- **Tech:** *Next.js, AWS, TypeScript, Python.*

The HALO Trust

Technology Intern

Dumfries, UK

Dec 2021 – Apr 2023

- **Geospatial AI Pipeline:** Conducted Remote Non-Technical Survey of Tarin Kowt, generating **1,500+** labels to calibrate **Mask R-CNN** models.
- **Operational Impact:** Produced baseline “Digital Twins” of hostile zones, quantifying risk for **12,000+ beneficiaries** and supporting **AHF funding** proposals.

PROJECTS

AI Conversational Placement Test | [View Case Study](#)

Solo Project

Pilot

2024 – Present

- **Operational Efficiency:** Compressed the student placement lifecycle from **2 weeks to 30 minutes** by replacing manual Zoom interviews with an on-demand AI assessment, enabling immediate class availability checks.
- **Assessment Flow System:** Translated expert examiner methodology into a deterministic assessment flow (Ask → Hint → Explicit), ensuring standardised evaluation across all candidates.
- **Multi-Tenant Architecture:** Engineered a platform supporting dynamic subdomains (***.languageetest.net**) using Next.js middleware and custom authentication to support distinct organisations.
- **Custom Evaluation Framework:** Built a custom eval suite with **~600 annotated samples**, optimising the multi-stage verification architecture to achieve **70% strict recall** on grammar pattern extraction.
- **System Reliability:** Built a custom version-control system for test configurations, enabling visual diffs, deployments, and atomic rollbacks.
- **Tech:** *Next.js, Cloudflare Workers, AWS, TypeScript, Vercel AI SDK (Gemini).*

ADDITIONAL INFORMATION

- **Languages:** Native English and Japanese speaker
- **Driving Licenses:** Full UK and Japan licenses